

Bulwark 1475 Prompt

You are an expert 2D game developer and front-end engineer who specializes in 16-bit Super Nintendo–style games.

TASK

Build a fully playable 2D 16-bit-style browser game called **“Bulwark 1475”** AND a matching **one-page marketing website** in a single project.

The game is inspired by early-1990s SNES arcade strategy/puzzle/shoot-’em-up games: top-down, chunky pixel art, fast rounds, and a tight gameplay loop built around three repeating phases:

1. Home castle selection (at start of each level).
2. Cannon Placement.
3. Combat (Battle).
4. Building (Repair/Expand).

...then repeat for multiple rounds and levels.

TECH & OUTPUT FORMAT

Use only vanilla **HTML, CSS, and JavaScript**. No frameworks, no build tools, no external CDNs. Everything must run as a static site by simply opening `index.html` in a browser.

Return **only code**, no explanations. Structure your answer as three clearly marked files:

1. `<!-- index.html -->`
2. `<!-- styles.css -->`
3. `<!-- game.js -->`

The content of each file should immediately follow its comment marker. Make sure `index.html` correctly references `styles.css` and `game.js`.

GAME OVERVIEW

Bulwark 1475 is a **top-down SNES-era castle defense game**. You command medieval island fortresses under siege from a navy or rival fort across a river / canyon / indestructible wall. Each round flows:

- **Home Castle Selection (only at start of each level)**

- **** Cannon Placement Phase **** – place cannons inside enclosed castles.
- **** Combat Phase **** – fire slowly moving cannonballs at enemy ships and forts.
- **** Building Phase **** – repair and extend walls using Tetris-like wall pieces.

If, after a Building Phase, ****no castle is fully enclosed by walls****, the game ends (you lose). If you survive all levels and rounds, you win, triggering a fun “walk the plank” execution of the losing commander.

CAMERA & VISUAL STYLE

- Camera: ****Bird’s-eye top-down view.****
- Environment:
 - Island with green grass/earth (slightly checkerboard or tiled).
 - Edges bordered by blue water.
 - Optional patches of sand/beach near shoreline.
 - A central river / canyon / indestructible wall separating player from enemy.
- Castles:
 - Small, chunky pixel-art castles.
 - Player’s home castle: blue-tinted stone.
 - Neutral or enemy castles: gray/tan.
- Walls:
 - Built from rectangular tile segments.
 - Gaps must be visually obvious (so players can see breaches).
- Color Coding:
 - Player cannons & UI highlights: blue.
 - Enemy cannons/ships: red/brown/dark.
- Ships:
 - Simple SNES-style galleons / warships, bobbing gently on the water.
- UI:
 - Stone/brick-themed HUD frame for:

- Level, round, phase name.
- Timer for current phase.
- Score.
- Number of enclosed castles.
- Number of cannons available.
- Animations:
 - Cannonballs: small dark circles that travel slowly, ideally with a slight arc.
 - Explosions: expanding circles or simple sprite flashes.
 - Ships bob slightly as they move.
- Final Victory:
 - After beating all levels, show a fun “walk the plank” animation:
 - Losing commander avatar shuffles along a wooden plank off a ship into the sea.
- Cartoonish, not graphic.

AUDIO (LIGHTWEIGHT)

- Use either `<audio>` tags or Web Audio API for:
 - Cannon fire.
 - Explosions.
 - Optional: short drum-loop background track per phase.
 - If full sound is too heavy, provide:
 - Simple audio hooks (`playCannonSound()`, `playExplosionSound()`, etc.)
 - A minimal example of loading/playing a sound.

CORE STRUCTURE: LEVELS, ROUNDS, PHASES

Levels & Rounds

- **5 Levels**, each a different island/map layout.
- Each level has **3 Rounds**.
- For each new level:

- Show a **Level Intro** screen.
- Player chooses a home castle.
- That chosen castle is instantly surrounded by a prebuilt, fully enclosed wall with internal space for that level's starting cannons (no building at the very start of a level).
- To win the entire game:
 - Survive all 5 levels × 3 rounds without losing all enclosed castles.
- To lose:
 - At the end of a Building Phase, **if no castle remains fully enclosed**, game over.

Difficulty should **scale per level**, for example:

- More and/or faster enemy ships.
- Longer or more intense combat.
- Stronger enemy cannons (more wall damage per hit).
- Possibly player cannons can be destroyed in later levels.

Phase Timing (per Round)

- **Home Castle Selection (per level):**
 - Player has **15 seconds** to pick a home castle from 5 available.
- **Cannon Placement Phase:**
 - Duration: **15 seconds**.
- **Combat Phase:**
 - Duration: **25 seconds**.
- **Building Phase:**
 - Duration: **20 seconds**.
- Between phases, show a **3-second title splash screen** that pauses the action and announces the next phase.

EXACT PHASE FLOW (PER ROUND)

0. Title Screen & New Game

- On page load, show a SNES-style **Title Screen**:

- Game title: “Bulwark 1475”.
- Pixel-art style logo text.
- Buttons:
 - “Start Game”
 - “How to Play” (scrolls to instructions section on the web page or opens an overlay).
- When “Start Game” is pressed:
 - Transition to **Level 1 Home Castle Selection**.

1. Home Castle Selection (per Level)

- Show a splash text overlay:
 - “Choose Your Home Castle” (3 seconds).
- Island map shows **5 castles** distributed around the island.
- Player has **15 seconds** to hover/click to choose one:
 - On hover: highlight castle with outline/glow.
 - On click:
 - This becomes the **home castle** for that level.
- Immediately auto-generate a **full wall enclosure** around the chosen castle:
 - Continuous loop of wall tiles.
 - Enough interior tiles to place the initial number of cannons.
- Once selected (or time expires and one is chosen automatically), go to **Cannon Placement**.

2. Cannon Placement Phase

- Show a **Cannon Placement title card**:
 - “Place Your Cannons” for 3 seconds.
- Duration: **15 seconds**.
- Available cannons:
 - **Home castle**: provides **2 cannons**.

- ****Each additional fully enclosed castle**** (own castles from previous rounds): ****+1 cannon****.

- UI shows:

- “Cannons remaining: X”.

- Phase timer countdown.

- Player interaction:

- Use mouse to move a ghost/preview cannon sprite over valid tiles.

- Click to place.

- Cannons ****must**** be placed:

- On tiles inside walls of ****fully enclosed castles**** they own.

- Not overlapping walls, other cannons, castles, or outside.

- After all cannons are placed or time runs out:

- Move to ****Combat Phase****.

3. Combat Phase

- Show a ****Battle title card****:

- “Get Ready for Battle!” for 3 seconds.

- Duration: ****25 seconds****.

- View:

- Player’s island and castles at one side (bottom or left).

- Enemy ships and/or enemy fort/castles across a river, canyon, or indestructible wall.

- Enemy Attack:

- Ships spawn at the water edge and sail toward firing positions.

- Enemy cannons (on ships or enemy forts) periodically shoot at player walls.

- Each hit on a wall tile:

- Damages or destroys that tile, creating a gap.

- Player Controls:

- **Active cannon** is highlighted with a crosshair.
- Keyboard controls (example mapping):
- `Q` / `E` : cycle between cannons.
- Arrow keys or A/D: adjust firing angle.
- Spacebar or Left Mouse: fire.
- Cannonballs:
- Slow-moving projectiles.
- Player must **lead shots** to hit moving ships.
- End of Combat Phase:
- Either when the **25-second timer** ends, or when all ships are destroyed (you still transition to Building Phase).
- Damage to walls persists into Building Phase.

4. Building Phase

- Show **Building title card**:
- “Rebuild Your Walls” for 3 seconds.
- Duration: **20 seconds**.
- Goal:
- Repair breaches around your castles.
- Optionally expand walls to **enclose additional castles**, gaining more cannons next round.
- Mechanics:
- Player is given a **sequence of random wall pieces**, Tetris-like, each a small polyomino shape.
- Current piece appears near side or top of the map.
- Controls example:
- Arrow keys: move piece on grid.
- `Z` / `X` or `R` : rotate.

- Spacebar or Enter: place.
- Rules:
- Pieces must snap to grid.
- Cannot overlap existing walls, castles, cannons, ships or non-buildable tiles (water, canyon).
- Pieces can create “overhangs” and awkward shapes, sometimes leaving small gaps that future pieces can’t fill.
- At **end of Building Phase**:
- Run an **enclosure check**:
- For each player-owned castle, check if it is completely surrounded by contiguous wall tiles (no leaks).
- If **zero castles** are fully enclosed:
- Trigger **Game Over** screen.
- If at least one castle is enclosed:
- Update:
- Number of enclosed castles.
- Cannons for next round.
- If 3 rounds completed at this level and player still has at least one enclosed castle:
- Progress to **next level** (new map).
- Otherwise, go back to **Cannon Placement** for the next round.

5. Game Over & Victory

- **Game Over**:
- If at any end-of-Building resolution step, player has zero enclosed castles:
- Show “Castle Fallen” screen.
- Display round, level, final score.
- Option to “Retry Level” or “Back to Title”.
- **Final Victory** (after Level 5, Round 3 successfully finishes):

- Show a celebratory screen.
- Then play a short “walk the plank” scene:
- Simple animation: enemy commander sprite walks onto a plank, hesitates, falls into water with a splash SFX.
- Show final score and prompt to play again or return to title.

SCORING SYSTEM

- Award points for:
- Destroying ships.
- Damaging/destroying enemy walls.
- Fully enclosing additional castles during Building.
- Completing a round without any wall breaches.
- Bonus:
- Extra bonus for having multiple enclosed castles at the end of a round.
- Display score prominently in HUD at all times.

GAME STATE & ENGINEERING DETAILS

Implement a simple game engine with clear state management:

- Core objects:
- ``gameState`` (global: title, playing, gameOver, victory).
- ``levelState`` (current level, current round, difficulty settings).
- ``phaseState`` (current phase: SELECT_HOME, PLACE_CANNONS, COMBAT, BUILD, SPLASH).
- ``player`` (castles, cannons, score).
- ``enemy`` (ships, enemy forts, AI parameters).
- ``tiles`` / ``map`` (terrain grid, walls, castles).
- Use `**requestAnimationFrame**`:
- A main loop ``gameLoop(timestamp)`` that:
- Calculates ``dt`` (delta time).

- Calls ``update(dt)`` and ``render(ctx)``.
- Phase handler functions:
 - ``startLevel(levelIndex)``, ``startRound()``.
 - ``startPhase(phase)``, ``updateSelectHome(dt)``, ``updateCannonPlacement(dt)``, ``updateCombat(dt)``, ``updateBuilding(dt)``.
 - ``renderSelectHome(ctx)``, ``renderCannonPlacement(ctx)``, etc.
- Wall and enclosure logic:
 - Store walls as a 2D grid or list of wall tiles.
 - For enclosure, you can use a simple region/flood fill from the castle outward; if you can't reach the map edge without passing a wall, it's enclosed.

CONTROLS SUMMARY (SHOW IN-GAME & ON WEBSITE)

- Mouse:
 - Click castles to select.
 - Click inside enclosed areas to place cannons.
- Keyboard:
 - Arrow keys / WASD: move cursor / move wall piece / adjust selection.
 - Q / E: switch active cannon.
 - Z / X or R: rotate wall piece.
 - Spacebar / Enter: fire cannon / place wall piece / confirm.
 - Esc: pause / open pause menu.

CANVAS & RESPONSIVENESS

- Single ``<canvas>`` element for the game.
- Keep internal resolution in a **pixel perfect** SNES-ish ratio (e.g., 256×224 or a 4:3 variant).
- Use CSS to scale the canvas up on larger screens while preserving aspect ratio.
- Ensure UI and text remain legible on laptops and tablets.

ONE-PAGE WEBSITE DESIGN (AROUND THE GAME)

The `index.html` should function as both:

1. The container for the ``<canvas>` game.
2. A simple **one-page marketing/landing site**.

Layout:

- **Hero/Game Section (top of page)**:
 - Game title + tagline:
 - Title: "Bulwark 1475".
 - Tagline: "Build the walls. Place the cannons. Survive the siege."
 - Prominent **"Play Now"** button that focuses the game canvas and starts the Title Screen / Game.
 - The canvas centered with a decorative frame (stone border using CSS).
- **Sections below hero**:
 1. **About the Game**
 - Short paragraph describing the medieval island fortress under siege, SNES-era visuals, and the core idea of alternating battle and building phases.
 2. **How It Plays (The Three Phases)**
 - Brief explanation of:
 - Home Castle Selection.
 - Cannon Placement.
 - Combat.
 - Building.
 3. **Levels & Difficulty**
 - Explain there are 5 levels with 3 rounds each.
 - Mention escalating difficulty (more ships, faster shots, stronger cannons).
 4. **Controls**
 - Clean table or bullet list of key bindings (mouse + keyboard).
 5. **Legacy & Inspiration**

- Short nostalgic blurb referencing 1990s arcade / SNES castle-defense vibes.

6. **Credits**

- Placeholder developer, artist, sound, year.

Styling:

- Background:

- Dark, subtle textured gradient or repeating pattern hinting at stone or parchment (pure CSS).

- Fonts:

- Use system-safe fonts but style them to feel “chunky”: bold, letter-spacing, drop shadows for headings.

- Colors:

- Blues, golds, and grays that match the game’s HUD and castles.

- Make the layout responsive:

- On wide screens, game + info can be side-by-side.

- On narrow screens, stack sections vertically with the game near the top.

IMPLEMENTATION REQUIREMENTS

- No external assets or CDNs required. If you want to prepare for future sprite or sound additions, design code so spritesheets/sounds can be swapped in easily later.

- Ensure that:

- Opening `index.html` in a browser just works.

- The game can be started via the “Play Now” or “Start Game” button.

- Include helpful inline comments in `game.js` describing:

- Main game loop.

- Phase state machine.

- Map/wall/enclosure logic.

- Input management (keyboard + mouse).

FINAL INSTRUCTION

Generate all three files now—`index.html`, `styles.css`, and `game.js`—as complete, ready-to-run code, formatted exactly as:

```
`<!-- index.html -->`
```

```
...html...
```

```
`<!-- styles.css -->`
```

```
...css...
```

```
`<!-- game.js -->`
```

```
...javascript...
```

No additional explanations or commentary, just the code.